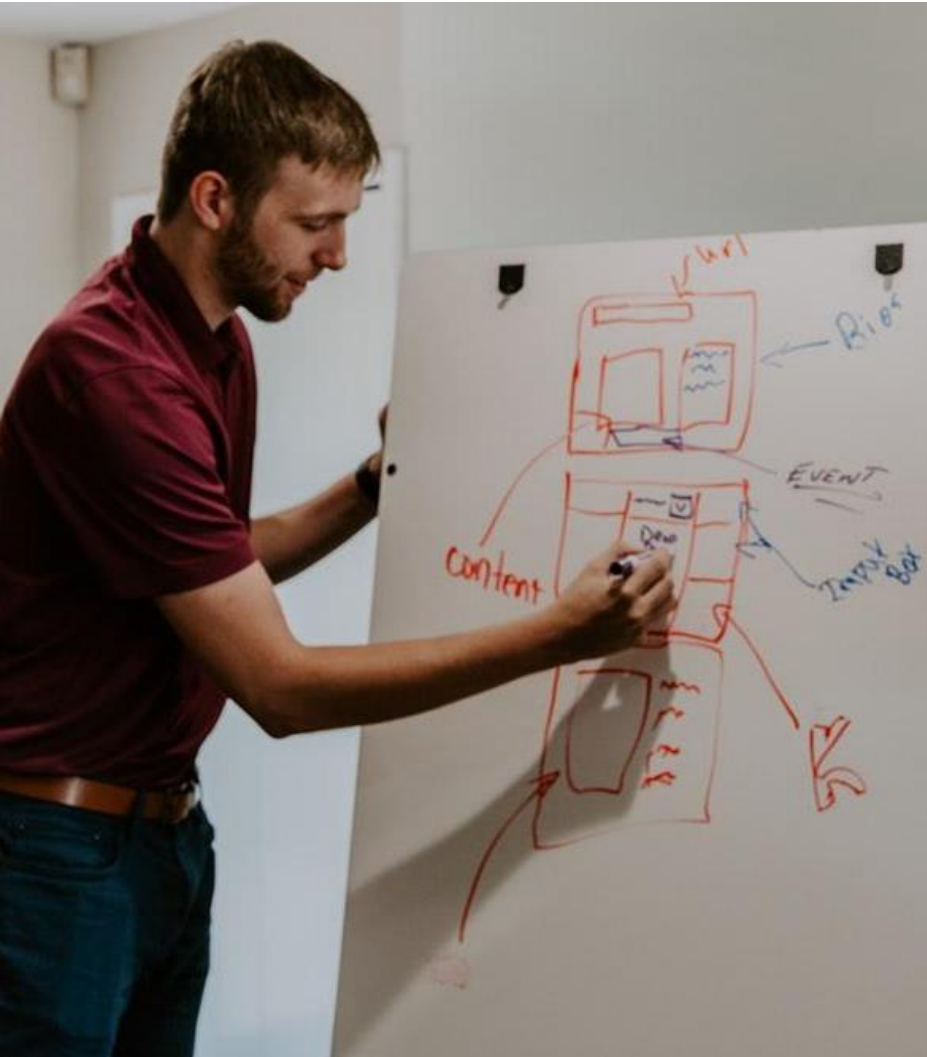


Use Case Description

Team Teaching OOAD

Informatics Engineering
Information Technology Departement
Politeknik Negeri Malang

Use Case Description



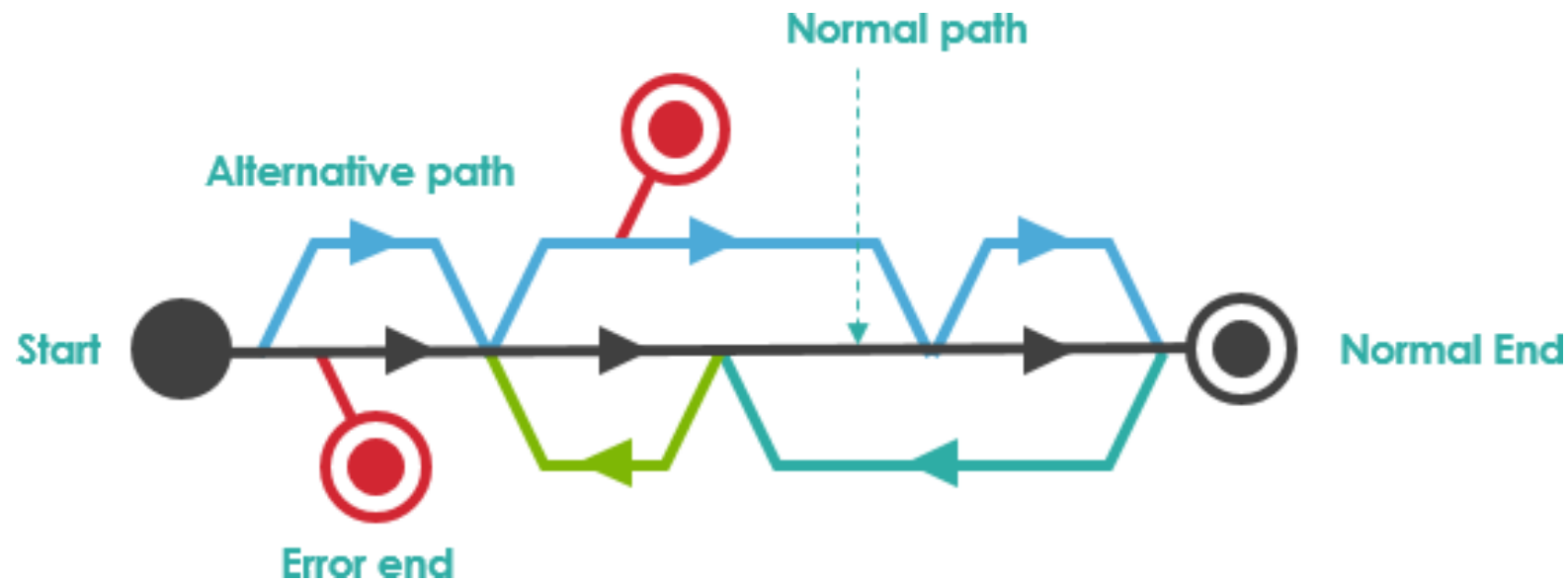
Use case description contains all the information to build structural and behavioral diagrams.

Expresses information in a less-formal way.

Has 3 basic parts:

- Overview information
- Relationship
- Flow of events

Use Case Description Illustration



Taken from visual-paradigm.com

Use Case Diagram Anatomy

Examples

Use Case Diagram – Overview Information (1)



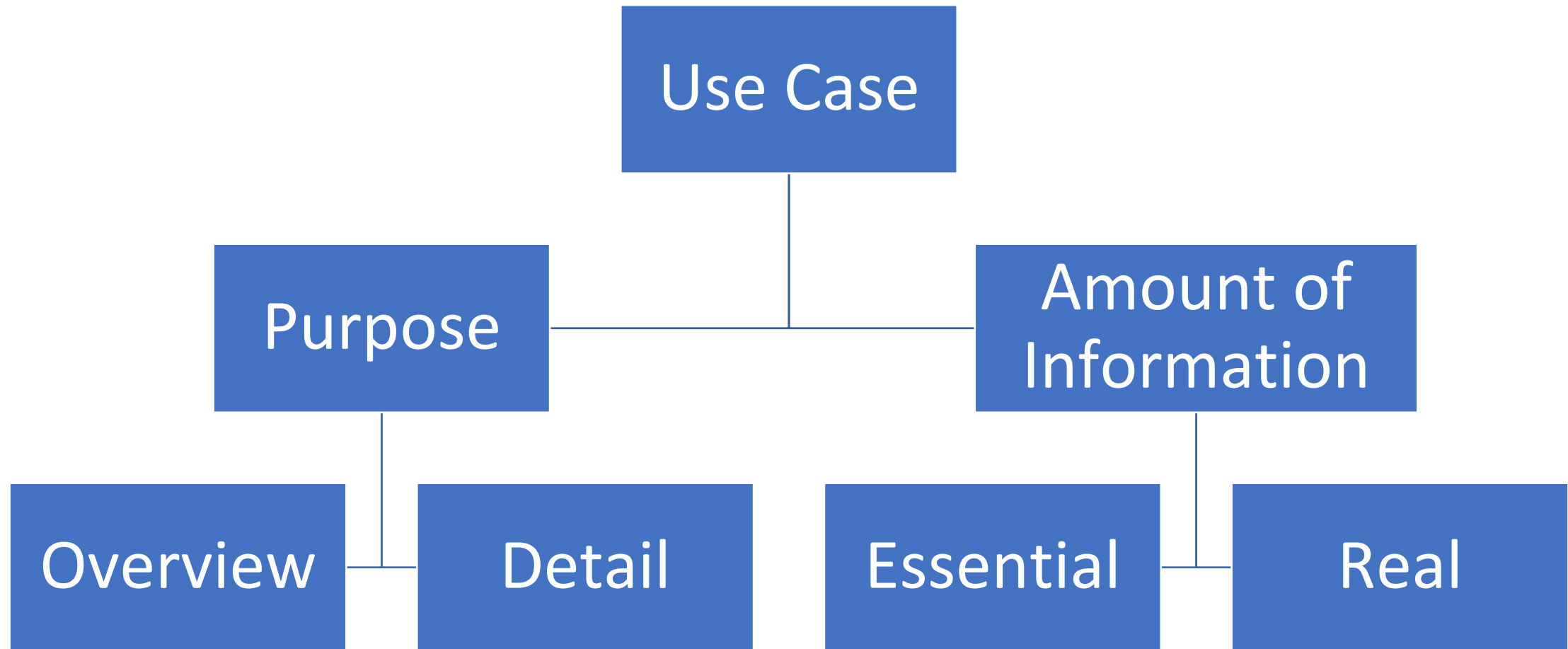
Use Case Name: Make Old Patient Appointment	ID: UC002	Importance Level: Hight
Primary Actor: Old Patient	Use Case Type: Detail, Essential	
Stakeholder and Interest: Old patient – wants to make, change, or cancel an appointment Doctor – wants to ensure patient’s needs are met in a timely manner		
Brief Description: This use case describes how we make an appointment as well as changing or cancelling an appointment for a previously seen patient		
Trigger: Patient calls and ask for a new appointment or ask to cancel or change an existing appointment		
Type: External		

Use Case Diagram – Overview Information (2)



- Use case name : Frasa verb-noun
- Use case ID : unique ID for every use case
- **Use case type** : type of use case (overview/detail and essential/real)
- Primary actor : Actor that triggers the use case
- Brief description : Generally in the form of a single sentence that illustrates the description of the use case
- Importance level : Priority Level in Use Case (high, medium, low)
- Stakeholder : Anyone related to Use Case. Always include Primary Actor.
- **Trigger** : The trigger that causes the use case begins.
 - External : Ex: Patient create an appointment
 - Temporal : Internal trigger or time based. Ex : generate monthly report

Use Case Types



Use Case Type: Purpose

Overview

Used to enable the analyst and user to agree on high-level overview of the requirements

Detail

All information needed for the use case. It can be based on the activities in the activity diagram

Use Case Type: Amount of Information



Essential

Describes only the minimum essential issues necessary to understand the required functionality

For example:

Essential > Patient want to make an appointment. The receptionist should attempt to match the patients desired appointment with the available times

Real

Describes specific set of steps

Real > Look on the available time using calendar and so on, more technical.

Use Case Trigger

- External Trigger: an event or condition **outside the system** that triggers actions or processes in the system.
- Temporal Trigger: triggers **based on time**.

Use Case Diagram – Relationships



Relationship:

Association: Old patient

Include:

Extend: Update Patient Information

Generalization: Manage Appointment

Use Case Diagram – Flow of Events (1)

Normal Flow of Event:

1. The Patient contacts the office regarding an appointment
2. The Patient provides the Receptionist with his or her name and address
3. If the Patient's information has changed, execute the **Update Patient Information**
4. If the Patient's payment arrangement has changed, execute the **Make Payments Arrangements** use case
5. The Receptionist ask Patient if he/she would like to make a new appointment, cancel an existing appointment, or change an existing appointment
 - a. If the Patient wants to make a new appointment, the S-1: new appointment sub flow is performed.
 - b. If the Patient wants to cancel an existing appointment, the S-2: cancel appointment sub flow is performed.
 - c. If the Patient wants to change an existing appointment, the S-3: change appointment sub flow is performed.
6. The Receptionist provides the result of the transaction to the Patient

Sub Flows:

S-1: New Appointment

1. The Receptionist asks the Patient for possible appointment times.
2. The Receptionist matches the Patient's desired appointment times with available dates and times and schedules the new appointment

S-2: Cancel Appointment

1. The Receptionist ask the Patient for the old appointment.
2. The Receptionist finds the current appointment in the appointment file and cancel it.

S-3: Change Appointment

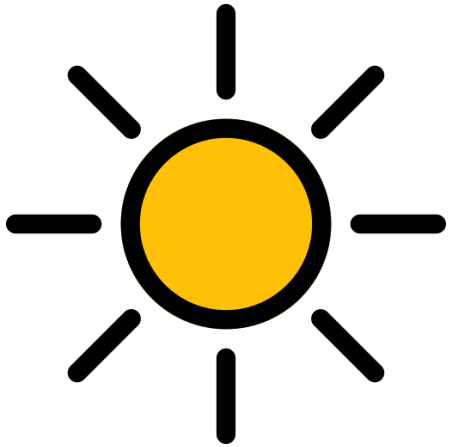
1. The Receptionist performs the S-2: cancel appointment sub flow.
2. The Receptionist performs the S-1: new appointment sub flow.

Alternate/Exceptional Flow:

S-1, 2a1: The Receptionist proposes some alternative appointment times based on what is available in the appointment schedule.

S-1, 2a2: The Patient choose one of the proposed times or decides not to make an appointment

Use Case Diagram – Flow of Events (2)



Normal (Basic) flow of event > Steps that **normally** are executed in a use case

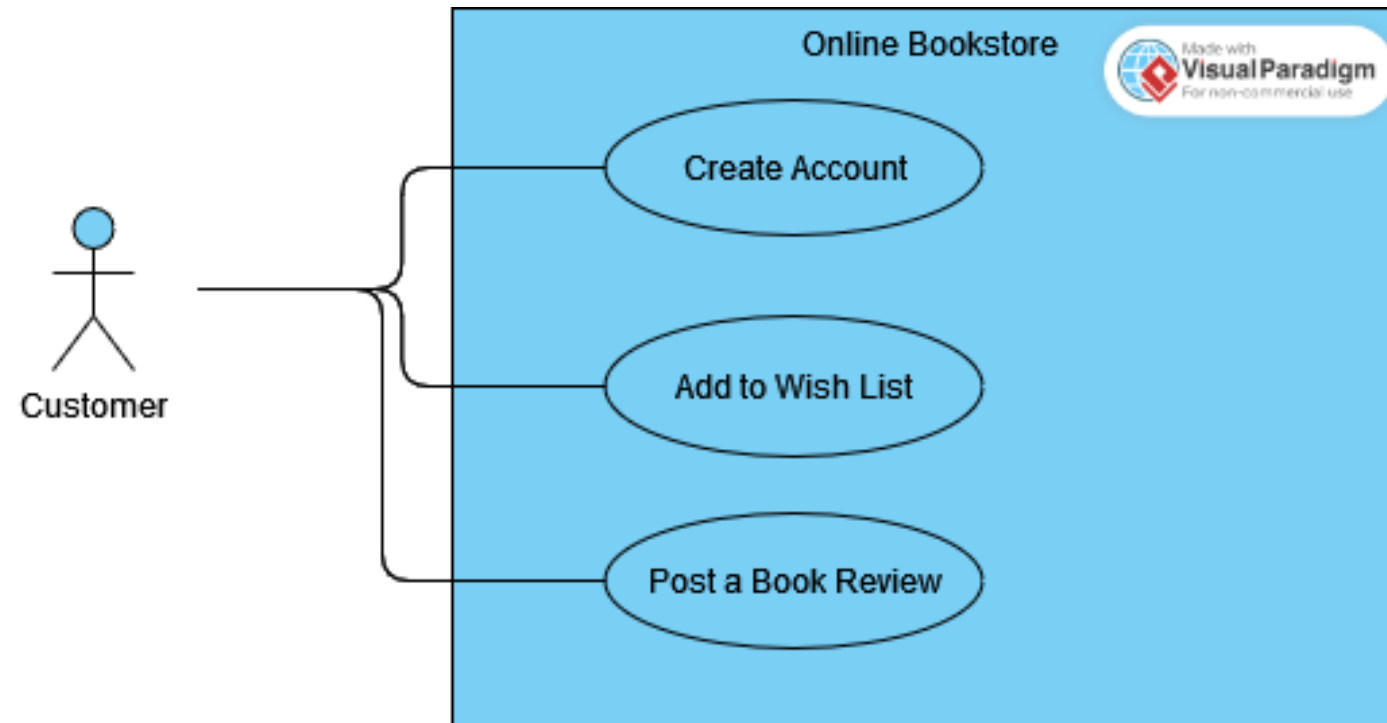


Alternative or **exceptional** flows > Ones that do happen but are **not considered to be the norm**

Back to our online bookstore

Refine, refine, and refine

Online Bookstore: The Use Case Diagram



Online Bookstore: The Use Cases

- The **customer** enter id, name, address, email, and password, to **create** a **customer account** and then click create account button. The system looks up the **master account list**. If the customer doesn't exist, the system will add the **customer account** to **master account list**.
- The **customer** select a **book** and **adds** it to his **wish list**. The system display the updated wish list page and display the customer's **shopping chart**.
- The **customer** able to **post** a **book review**. The system will show the reviews on the book detail page.

The Use Case Description of Create Account (1)

Use Case Name: Create Account	ID: UC001	Importance Level: High
Primary Actor: Customer	Use Case Type: Detail, Essential	
Stakeholder and Interest: Customer – wants to create a new account in the online bookstore web application		
Brief Description: This use case describes how customer make an account		
Trigger: Customer click on register button at the online bookstore homepage		
Type: External		
Relationship:		

The Use Case Description of Create Account (2)

Normal Flow of Event:

1. The customer opens the online bookstore
2. The customer clicks on register button
3. The customer provides id, name, address, email, and password
4. The customer clicks create account button
5. The system gets the customer information from point 3
6. The system checks master account list
7. If the account doesn't exist, the system creates a new account for the customer and display login page

Sub Flows:

Alternate/Exceptional Flow:

If account exist, the system displays an error message and redisplay the create account form page

